BNY Mellon Chennai

About the company:

The Bank of New York Mellon Corporation, commonly known as BNY Mellon, is an American investment banking services holding company headquartered in New York City. BNY Mellon was formed from the merger of The Bank of New York and the Mellon Financial Corporation in 2007. It is the world's largest custodian bank and securities services company. The bank's primary functions are managing and servicing the investments of institutions and high-net-worth individuals. Its two primary businesses are Investment Services and Investment Management. The bank's clients include 80 percent of Fortune 500 companies. The Mumbai Representative Office of The Bank of New York Mellon was set up in 1983 and continues to serve as BNY Mellon's main liaison point for Indian client base. In 2013, Pershing India Operational Services Private Limited (Pershing India) was established in Chennai. Currently, company have over 12,000 employees across our three locations: Mumbai, Pune and Chennai.

My Interview Experience:

No of Rounds: 3

1) Online Test (Coding Round)
2) Technical Interview
3) Technical + HR Interview

Round 1:

Coding Round:

- Conducted in Hacker rank Website
- Test Duration is 90 mins
- Camera On
- 1 easy, 2 medium and 1 hard question.
- Only function not full code.

Coding questions:

First Question (Easy):

Given an array, find the efficiency of an array. To find the efficiency choose a pair such that all pair sum should be equal and then eff+=pair.first+pair.second. Efficiency should be maximum.

Test case 1:

Input:

N=6

A=[4,3,2,3,1,5]

Output:

22

Explanation:

Eff=1*5+2*4+3*3 (Here 1+5==2+4==3+3)

Test case 2:

Input:

N=[1,2,2,7,2,4]

Output

-1

Explanation:

No pairs can be found such that sum is equal in all pairs.

Topics: Array, Sorting.

<u>Second question (Medium):</u>

Given two strings such that convert one string to another string with only two operations given applying any number of times.

Return 1 if possible else 0.

Two operations:

1) You can swap str[i],str[i+1]
2) Substring of k (k>1) size with same elements (ASCII Value=K) in that substring can be converted to next elements (ASCII Value=K+1). Example: “aaadcd” to “bbbdcd”. Here k=3 and with ‘a’ same element in that substring is converted to next element ‘b’. Note: ‘z’ can’t be converted to next element.

Test case 1:

Input:

”caa”

“bbc”

Output:

1

Explanation:

Operations:

Swap s1[0],s1[1] => “aca”

Swap s1[1],s1[2] => “aac”

Then k=2 from 0-1 index with a element to next b element => “bbc” ==str2

So output 1:

Test case 2:

Input:

”abc”

“bcd”

Output:

0

Explanation:

We can’t convert str1 to str2 with those two operations.

Topics: Strings, HashMap.

<u>Third Question (Medium):</u>

Given a 2D matrix with 0 and 1 values in it. 0 -> you can go and 1 -> you can’t go. K value given.

K -> you can jump at most k steps in horizontal and vertical direction. We have to find minimum no of steps required to go from (0,0) to (N-1, M-1). If not possible return -1.

Test case 1:

Input:

0 0 1

0 0 0

N=2

M=3

K=3

Output:

2

Explanation:

(0,0)[k=1]->(1,0)[k=2]->(3,0). There is no obstacle in the path.

Test case 2:

Input:

0 1 0

1 0 0

N=2

M=3

K=3

Output:

-1

Explanation:

We can’t move to any cell from (0,0).

Topics: DFS, DP, Backtracking.

<u>Forth Question (Hard Problem):</u>

Given an array. You have to apply operations in it

Operations:

1) 1 v w => change vth element to w
2) 2 v v => change every element in that array which are less than v to v.

These two operations are given in sequences. You have to apply and return the modified array.

Test case 1:

Input:

[7,2,5]

Operations:

2 6 6

1 3 9

2 8 8

Output:

[8,8,9]

Explanation:

2 6 6 => [7,6,6]

1 3 9 => [7,6,9]

2 8 8 => [8,8,9]

Topics: Array, 2D Array, Sorting, Searching, Priority Queue.

For me 1st question 15/15, 2nd question 5/15, 3rd question 7/15, 4th question – 10/15 test cases passed.

25 students shortlisted for round 2 out of 150-200 students. I was one of them.

<u>Round 2:</u>

Technical Interview:

The interview was conducted in CUIC. Interview conducted in physical mode. First, he read my resume and after that he started to ask questions. Interview took around 50 mins

Questions:

1) Project Related questions:

a) Tell about your DBMS project?
b) Draw tables.
c) Tell how they are connected (primary key and foreign key)?
d) He gave me some suggestions in my tables.
e) How will 1000 hotel details will be displayed in that website (My Project is hotel booking system)?
f) How did you implement ACID properties in your project?

2) Technical questions:
   a) Uses of operating systems
   b) Deadlocks
   c) Interpreter and Compiler
   d) Why we are using binary systems in computers
   e) Triggers and procedures differences
   f) Condition for UNION in DBMS
   g) Left, Right and Full Join
   h) Transactions
   i) ACID Properties Transactions
   j) Commit, Rollback {If I didn't commit and server crashed, what will happen}.
3) DSA question:
   Given an array with characters. To store the frequency of characters ('A'-'Z') .
   Constraints:
   a) Only Array (Not HashMap etc)
   b) Time Complexity O(N)

   Solution:

```
freq=[]
for i in range(0,26,1):
    freq.append(0)
arr="NAVANEETH"
for i in range(0,len(arr),1):
    freq[arr[i]-65]+=1
print(freq)
```

   Topics: Array, ASCII values.

4) General Questions:
   a) Which one you would prefer frontend or backend?
   b) How will you debug the code?
   c) Which IDE you use?
   d) Why did you choose CEG instead of NIT's?
5) Puzzle
   https://www.geeksforgeeks.org/puzzle-bag-of-coins/

Answer the questions confidently. Overall, the interview went very well and I was shortlisted for the next round.

Round 3:

Technical and HR Interview:

First, she read my resume and after that she introduced herself and asked me to introduce myself.

This interview is combined like both technical and HR. She asked to design a project using C++. Interview went around 45 mins - 1 hour

Questions:

1) What project you will choose (I told Chess Game)
2) Write some functions related to that game (No need to write full function only name and tell what that function do. I did use OOPS (classes))
3) What is OOPS, explain in depth
4) Difference between Abstraction and Encapsulation.
5) How will you implement Abstraction and Encapsulation in your project.
6) What constraints you used in your project
7) How will you debug if you found any error
8) If you are submitting it to a professor, on that time your code is not working what will you do?
9) If professor asks you to do improvement in your project, what improvements you will do
10) If professor asks you to do this project as team or individual which one you will prefer
11) (I told team) What kind of team you will choose
12) Tell about team conflicts anything happened in your life
13) How will you guide your team
14) If anyone of teammate is not contributing what you will do
15) Tell something interesting about BNY Mellon
16) What will you do after graduation
17) Why did you learn C++ and python instead of java

At last, she asked, do you have any questions. I asked what tech stacks are used in summer internship.

16 students selected for internship. I was one of them.

Do check out past intern interview experiences from GFG and practice company specific question sets too.

<u>Points to remember:</u>

- Practice coding questions
- Prepare "Introduce yourself"
- Try to pass more testcases rather than concentrating only on one question in coding round.

<u>TOPICS TO BE FOCUSSED:</u>

- Data Structure
- Database management system
- Operating system
- Whatever you've put in your resume
- OOPS

"ALL THE BEST"

<u>About me:</u>

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